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Towards Smart Farming: A Real-Time Diagnosis System for Strawberry Foliar Diseases Using Deep Learning

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Abstract

Background: Developing an effective machine vision system is crucial to successfully deploying robotic inspection in open field conditions and controlled environments like greenhouses. Robotic arms with vision-based deep learning models offer an efficient, real-time, non-invasive crop monitoring solution. In agricultural settings, they enable consistent, automated inspection under varying conditions, reduce labor dependency, and support early disease detection, enhancing productivity and sustainability in precision farming. Although considerable progress has been made in computer vision-based approaches, significant challenges persist in developing models that reliably perform under the diverse and variable conditions encountered in real-world agricultural settings.

Method: Within the domain of precision agriculture, we introduce an advanced robotic system for the detection of plant diseases, utilizing an innovative model based on deep learning principles. This system introduces an algorithm for real-time analysis, called as Strawberry Leaf Disease Inspection (SLDI). The algorithm integrates the use of Receptive Guided Channel Attention *(RGCA)* alongside a Deep Context Aggregator *(DCA)*, designed to significantly improve the characterization and representation of feature sets, thereby enhancing the overall accuracy and efficiency of disease identification. To optimize the system performance and preserve real-time performance, a Multi-Scale Feature Fusion

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Module *(MSFF)* is proposed that facilitates a comprehensive multi-level representation, enabling the model to capture disease symptoms promptly. The SLDI algorithm is deployed on a robotic platform equipped with an RGB camera, enabling real-time, in-field inspection of strawberry crops.
Results: The proposed system is trained on two publicly available datasets, PlantDoc and PlantVillage. It attains a precision of 91.10% and a recall of 88.50%, while maintaining a real-time processing speed of 76.50 frames per second (fps). Experimental field inspection of strawberry studies demonstrates that the proposed model significantly outperforms existing approaches in accuracy and efficiency.

Keywords: Attention network, Computer vision, Guided attention, Strawberry leaf disease, Smart farming, Universal robot UR5

1 Introduction

The strawberry (Fragaria spp.) plant is widely recognized as a suitable fruit crop for cultivation in highland regions, particularly at elevations between 1000 and 1500 meters above sea level [11]. However, identifying specific diseases affecting strawberry plants remains a persistent challenge for farmers, particularly in large-scale or labor-constrained operations [30]. In this context, integrating robotic arms equipped with intelligent vision systems offers a compelling solution for automated plant health monitoring. These systems can be deployed in controlled environments (e.g., greenhouses) and open-field agriculture, enabling consistent, non-invasive inspection of plant foliage with high spatial and temporal precision. In real-world settings, robotic arms provide several advantages: they can operate continuously under varying conditions, minimize human labor and error, and enable early, localized detection of foliar diseases. This facilitates timely intervention, reduces chemical usage, and contributes to higher crop yield, improved quality, and sustainable farming practices. Traditionally, the inspection of strawberry foliar diseases has relied heavily on manual observation by experienced agricultural workers or plant pathologists. This approach typically involves visual assessment of leaf discoloration, spots, mold, or other physical symptoms indicative of disease [28]. While such methods have been widely practiced, they are time-consuming, labor-intensive, and highly subjective. The accuracy of manual inspection often depends on the inspector's expertise, and it may lead to inconsistent or delayed diagnoses, especially under conditions of large-scale farming or limited access to skilled personnel [7]. In recent years, Machine Learning (ML) techniques have been increasingly explored in agricultural diagnostics as an alternative to manual inspection. These methods typically rely on handcrafted features extracted from leaf images to train predictive models. However, they present several limitations: The manual feature extraction process requires domain expertise and often fails to capture plant disease's complex and varied patterns. Such models also lack generalization across different environments and are sensitive to variations in lighting, cultivar types, and

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disease stages. Their reliance on limited datasets and extensive preprocessing can hinder scalability, real-time performance, and overall reliability in practical applications [32]. These challenges have motivated a shift towards more advanced and automated solutions, particularly those based on deep learning architectures that can learn hierarchical features directly from raw image data, eliminating the need for manual feature engineering.

An Effective detection and management of strawberry leaf diseases are critical to minimizing agricultural losses and sustaining the long-term health of strawberry cultivation. Hence, visual feature extraction via deep learning effectively distinguishes various plant diseases [34, 54–56]. For example, Wu et al. [43] modified the ResNet model to classify diseases in strawberries effectively. Additionally, Bera et al. [3] developed the PND-Net by integrating Graph Neural Networks (GNNs) with Convolutional Neural Networks (CNNs), presenting better performance on the discrimination of numerous diseases. However, these approaches have architectural limitations in plant disease analysis. CNNs, with their fixed receptive fields and local filtering, often miss fine-grained or irregular lesion patterns and lose global context. GANs, while useful for data augmentation, may generate unrealistic textures and lack semantic control, which can compromise model reliability. Both architectures generally lack pixel-level localization, limiting their effectiveness in tasks requiring precise spatial interpretation. Moreover, Wang et al. [46] presented MatDet, a model tailor for determining tomato maturity under challenging conditions. Liu et al. [27] optimized YOLO's multi-scale feature extraction to improve tomato pest and disease assessment. Furthermore, Li et al.[26] leveraged deep transfer learning for pest and disease detection in tea plants. Meanwhile, Liu et al.,[28] employed spatiotemporal fusion attention techniques to improve multi-source vegetable disease identification. The detection models, while fast, often struggle with detecting small or overlapping disease regions due to coarse grid-based predictions. Though effective in capturing dynamic patterns, Spatiotemporal attention networks can be computationally intensive and less suited for real-time agricultural deployment.

This study presents an efficient deep learning framework, *Strawberry Leaf Disease Inspection (SLDI)*, for real-time identification of strawberry foliar diseases. The SLDI pipeline integrates a YOLO-NAS object detection backbone with a Transformer-based encoder to combine rapid and precise localization with enhanced contextual understanding of disease patterns. To further improve feature representation, the architecture incorporates Receptive Guided Channel Attention (RGCA), Deep Context Aggregator (DCA), and a novel Multi-Scale Feature Fusion Module (MSFF). These complementary features are fused through dedicated modules (M1, M2, and M3) as illustrated in Fig. 1, ensuring robust detection under challenging conditions such as occlusions, small-scale lesions, and variable environments. The trained SLDI model has been deployed on a robotic arm equipped with an RGB camera, enabling automated, contactless, and real-time inspection of strawberry plants in agricultural settings. This integration of deep learning and robotics provides a scalable, sustainable solution for early disease detection and smart crop management.

The following are the main contributions of the current work:

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- An automated strawberry leaf disease inspection system is developed using a robotic arm integrated with a vision-based deep learning model. The system provides real-time, autonomous disease detection, aiming to improve accuracy, reduce manual effort, and minimize human error. Designed for adaptability to diverse field conditions, it enables large-scale crop health monitoring in greenhouses and open fields, supporting precision farming, reducing pesticide usage, and enhancing yield quality.
- We introduced two novel modules, Receptive Guided Channel Attention (*RGCA*) and Deep Context Aggregator (*DCA*), to enhance feature representation for disease detection. *RGCA* dynamically emphasizes informative channels based on receptive field variations, improving sensitivity to subtle disease patterns. In parallel, *DCA* captures rich multi-scale contextual dependencies, enabling better understanding of spatial and semantic relationships within complex leaf textures. Together, these modules enhance the model's ability to distinguish fine-grained disease features, generalize across varying image conditions, and reduce reliance on large-scale pre-training. When deployed on robotic or mobile platforms, *RGCA* and *DCA* support robust, real-time plant disease identification under diverse environmental conditions, advancing scalable smart farming solutions.
- We developed a novel fusion mechanism, Multi-Scale Feature Fusion (*MSFF*), to effectively integrate hierarchical features across different network stages. *MSFF* facilitates seamless flow between low-level spatial details and high-level semantic information, enabling the model to capture fine-grained disease patterns while preserving global context. By bridging shallow and deep feature representations, *MSFF* improves robustness against scale variation and visual complexity, leading to enhanced detection accuracy and feature continuity. When deployed in agricultural robotic systems, *MSFF* supports reliable, real-time disease identification under diverse leaf conditions, increasing adaptability and effectiveness in smart farming applications.
- To thoroughly evaluate the proposed model, we conducted comprehensive ablation studies and benchmark comparisons against several state-of-the-art plant disease detection methods. Experiments were performed on two widely used datasets, Doc and PlantVillage, to assess robustness and generalizability. The proposed model achieved a precision of 91.10% and a recall of 88.50%, demonstrating its ability to accurately detect and localize disease symptoms under diverse conditions. These results underscore its potential for real-world deployment in agricultural applications.

To comprehensively explore the topic, this manuscript adheres to the following structure: Section 2: represents the related literature on the detection of crop diseases, providing a comprehensive overview. Section 3 introduces a proposed approach for accurately detecting strawberry leaf disease using deep learning technique. Section 4 covered the technical details about the implementation, datasets, ablation study, comparative analysis with SOTA methods. Section 5 provides the detailed experimental results of the work. Section 6, presents the discussion, while Section 7 summarizes this research with future research directions.

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2 Related Work

Early and accurate recognition of plant diseases is essential for effective crop management and ensuring global food security. Based on an in-depth literature review, plant disease detection methods can be broadly categorized into CNN-based and transformer-based approaches. A comprehensive analysis of these methods is presented in the following subsections.

2.1 CNN-Based Object Detectors

Previous research has thoroughly investigated the agriculture scene analysis of strawberry plant diseases [21–23]. [36] utilized CNNs to recognize strawberry plants diseases and compared VGG-16, ResNet-50, and G-Net, with ResNet-50 to evaluate the highest accuracy. Another research study [14] utilized a diverse selection of crop leaves from the PlantVillage dataset and employed CNN architectures involving InceptionV3, InceptionResNetV2, MobileNetV2, and EfficientNetB0 to detect plant diseases. Similarly, [33] proposed a CNN algorithm based on MobileNetV3-Large and EfficientNet-B0 to detect three diseases of the strawberry leaf, where the Large network achieved an accuracy of 92.14%. These studies highlight the effectiveness of different CNN architectures in accurately detecting plant diseases, including those specific to strawberries.

In the current studies, researchers explored either directly or fine-tuned CNN architectures to identify plant diseases. For instance. [50] used and compared ResNet50, VGG16, and GoogLeNet to classify three strawberry diseases, where ResNet50 attained the highest precision of 99.60%. To identify four grape leaf diseases, [19] compared UnitedModel with VGGNet, GoogLeNet, DenseNet, and ResNet. They found that UnitedModel achieved an accuracy of around 99.17%. [2] achieved an identification precision of 99.39% for 39 classes of leaf disease of plants using a pre-trained EfficientNet-B5 model. [2] applied a pre-trained Inception-V3 model to identify six classes of plant disease, achieving an accuracy of 94.20%. They also created BR-CNN, incorporating a Binary Relevance (BR) multi-label learning algorithm with a lightweight NasNet to detect 10 plant diseases. However, these models are susceptible to over-fitting when applied to small target datasets or when they have many parameters. BR-CNN demonstrated an identification accuracy of 99.17%, surpassing VGGNet by 4.89%.

Despite the excellent accuracies attained with large-scale datasets, deep learning networks trained on a small number of annotated plant leaf disease images frequently perform unsatisfactorily due to the few-shot problem, specific to this domain. [45] introduced a model that combines VGGNet, GoogLeNet, and ResNet to find fine-grained assessment of the severity of bacterial spot diseases in peppers. This approach overcame the limitations of individual CNNs by combining the extracted features, attaining an optimal precision of 95.34%. [48] developed a model incorporating GoogLeNet and DCGAN to identify five common tomato diseases, achieving a precision of 94.33%. Although CNNs have shown remarkable success in detecting plant disease, however, they often grapple with challenges such as architectural complexity, high computational demands, and susceptibility to both under and over-fitting, mainly

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when dealing with limited data sets or complex disease patterns. However, conventional CNNs suffer from fixed receptive fields that limit multi-scale lesion detection and exhibit over-sensitivity to lighting and scale variations common in field conditions. These drawbacks motivate our inclusion of grouped attention mechanisms to dynamically adapt receptive fields.

2.2 Transformer-Based Object Detectors

In precision agriculture, transformer-based networks have emerged as a powerful alternative to Convolutional Neural Networks (CNNs), effectively leveraging the fusion of RGB, multispectral, and hyper-spectral data to enhance agricultural monitoring tasks. [42] introduced Agricultural Aerial Transformer (AAFormer), a model that employs data fusion and attention mechanisms to enhance the accuracy of crop classification and disease detection, outperforming traditional methods. AAFormer's scalability and efficiency mark a significant advancement in precision agriculture. [9] demonstrated that a hybrid model, integrating CNNs and Vision Transformers, yields superior accuracy in classifying tomato leaf diseases. [37] proposed an enhanced Vision Transformer model designed explicitly for plant disease recognition. The model optimizations focus on capturing intricate patterns and features in plant images, increasing accuracy and efficiency. [51] presented a novel approach that combined deep attention-based learning in multiple instances with Vision Transformers. This method enhances crop disease classification by focusing on relevant image regions and capturing complex patterns. [4] used a deep learning methodology with Vision Transformers to classify plant diseases automatically. By using the algorithm's ability to identify intricate patterns in visual data, it surpasses the effectiveness of conventional methods in recognizing plant diseases with a high degree of accuracy.

Vision Transformers (ViT) [10] has opened new avenues, overcoming a few limitations. Researchers are increasingly adopting advanced ViT architectures [25, 47] for dense object detection tasks. Jiang et al.,[18], for example, enhanced the YOLO model by integrating ViT, creating a transfer-learning framework for detecting cabbage diseases in field conditions. Similarly, Ye et al. [53] developed a tomato leaf disease recognition system by combining multi-scale features with transformer-based attention mechanisms. To improve crop disease detection comprehensively, it's crucial to integrate local and global feature extraction, leveraging the strengths of CNNs and transformers. Recent efforts have explored hybrid architectures that merge CNNs' robust local feature capture with transformers' global contextual understanding. Guo et al. [12, 39] demonstrated how integrating CNNs with transformers can overcome the performance limitations of CNNs alone, without extensive pre-training. Despite these advances, hybrid models have predominantly been explored in various domains involving image classification, detection, and segmentation [6, 31].

Several studies [13, 20] have explored the application of Multi-scale Vision Transformers and Swin Transformers for crop disease recognition. By using their ability to analyze images at multiple scales and capture fine-grained details, these models surpass traditional methods in both accuracy and performance. [53] introduced Tswin-F, a novel Transformer-based tomato leaf disease detection network. The network employs bilateral local attention and self-supervised learning to capture spatial image

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connections. Furthermore, it includes a Feature-Fuse Local Attention (FFLA) structure to address the problem of increasing attention distances in deep Transformer models. Tswin-F has shown high precision in 10-class and 13-class tomato leaf disease datasets. [29] introduced plant-based MobileViT (PMVT), a computationally efficient model for real-time plant disease detection based on MobileViT. PMVT is designed to be lightweight and attain high accuracy, which makes it appropriate for use on resource-constrained mobile devices. In another study by [37], The main goal was to enhance the ViT Base (B) transfer learning model for rice disease identification. The model's capacity to beat conventional CNN models like VGG, MobileNet, and EfficientNet was proved by the researchers via meticulous reconfiguration, layer augmentation, and hyperparameter optimization, especially when handling unbalanced datasets. [52] proposed a novel transformer block designed for the recognition of fine-grained plant disease. This block combines transformer architecture for global context modeling, soft split token embedding for capturing local details, and inception architecture with cross-channel feature learning to enrich feature representation. The model achieved state-of-the-art accuracy on multiple plant disease datasets. The downside of using only transformer detectors is their high computational cost and latency. They also frequently perform poorly when localizing small objects, a result of their global self-attention handling features at too coarse a level.

To address this, hybrid CNN-transformer approaches, such as *RGCA+DCA*, provide a good trade-off. Such methods successfully merge local feature sensitivity with global context, incurring minimal overhead.

2.3 Comparative Analysis

CNNs and ViTs address plant-health imagery from opposite ends of the modeling spectrum, each with well-documented trade-offs. CNNs exploit local convolutions and translation-equivariant pooling, making them parameter-efficient and effective on modest, class-imbalanced agricultural datasets. These same inductive biases, however, keep the effective receptive field largely local; capturing long-range context or lesions that span multiple scales typically requires auxiliary structures such as feature-pyramid networks or atrous spatial-pyramid pooling. Pooling layers further introduce a loss of spatial resolution, which can blur fine lesion boundaries. ViTs, by contrast, replace convolutions with self-attention over non-overlapping image patches, allowing global context aggregation from the first layer. This comes at a cost of self-attention scales quadratically with the number of tokens, inflating memory and compute demands, while the absence of strong locality biases means ViTs rely on substantial pre-training or large, diverse datasets to learn spatial regularities. Positional encoding must also be tuned carefully poor choices can degrade fine spatial detail. Finally, although ViTs can model multi-scale cues implicitly, practical deployments often add pyramid or windowed attention modules to curb computation. The technical limitations between these two networks are reported in the Table 1.

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Table 1 Architectural limitations of CNNs and Transformers for plant disease detection.

Aspect	CNNs	Transformers
Receptive Field	Effective receptive field remains localised, limiting long-range context modelling.	Self-attention captures global context but its quadratic complexity hinders high-resolution scalability.
Multi-scale Features	Fixed pooling hierarchies reduce adaptability to lesions of varying size.	No inherent multi-scale modelling; extra modules (e.g., pyramids) are required.
Spatial Awareness	Pooling gives translation invariance yet can blur precise lesion localisation.	Requires positional encoding; suboptimal choices lose fine spatial detail.
Parameter Efficiency	Convolutions are compact, but very deep CNNs risk vanishing gradients and over-fitting on small data.	High parameter count and memory footprint make training on small datasets difficult.
Inductive Bias	Strong locality/shift biases benefit small data but restrict global pattern learning.	Minimal inductive bias increases dependence on large datasets to learn spatial relations.

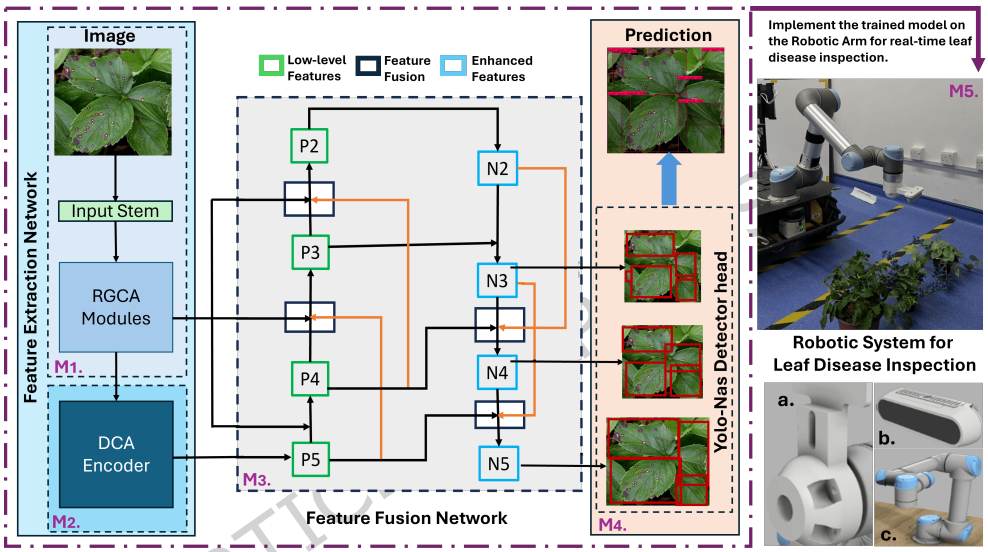


Fig. 1 **The Strawberry Leaf Diseases Inspection (SLDI) architecture is deployed on a robotic system for practical, real-world applications. This implementation involved (M1) *RGCA*, (M2) *DCA* modules for discriminative features extraction, and a multi-scale object detection layer is applied by using (M3) *YOLO-NAS*. Later, an optimized Pyramid Attention Network *PAN* and *MSFF* modules for feature fusion were implemented in the robot. The robotic system includes an end effector, a camera, and a manipulator, facilitating automated inspection of strawberry leaves for disease detection.**

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Table 2 Layer–wise specification of the proposed SLDI detector (Batch-Norm after every convolution; ReLU is the default activation except SiLU inside the YOLO-NAS head).

Stage	Layer / Block	Input (H×W×C)	Output (H×W×C)	Activation
Stem	Conv 7×7, stride 2 MaxPool 3×3, stride 2	640×640×3 320×320×64	320×320×64 160×160×64	ReLU —
ResNet-50 backbone				
Conv2_x	3 Bottleneck blocks	160×160×64	160×160×256	ReLU
RGCA-1	Region-Guided Ctx Attn.	160×160×256	160×160×256	ReLU
Conv3_x	4 Bottleneck blocks (s=2)	160×160×256	80×80×512	ReLU
DCA-1	Dilated Ctx Attn.	80×80×512	80×80×512	ReLU
Conv4_x	6 Bottleneck blocks (s=2)	80×80×512	40×40×1024	ReLU
RGCA-2	Region-Guided Ctx Attn.	40×40×1024	40×40×1024	ReLU
Conv5_x	3 Bottleneck blocks (s=2)	40×40×1024	20×20×2048	ReLU
MSFF	FPN top-down fusion	Conv3–5	P3 80×80×256, P4 40×40×256, P5 20×20×256	ReLU
DCA-2	Dilated Ctx Attn. on P5	20×20×256	20×20×256	ReLU
YOLO-NAS-S neck (ELAN C2f, SPPF, PAN-FPN)				
C2f stem	Conv–C2f blocks	320×320×64	320×320×128	SiLU
C2f-down-1	Down + C2f 6×	320×320×128	160×160×256	SiLU
C2f-down-2	Down + C2f 6×	160×160×256	80×80×512	SiLU
C2f-down-3	Down + C2f 3×	80×80×512	40×40×1024	SiLU
SPPF	Spatial pyramid (5,9,13)	40×40×1024	20×20×1024	SiLU
PAN-FPN	Up/concat + C2f 3×	20–80 feat. pyramid	P3 80×80×256, P4 40×40×512, P5 20×20×1024	SiLU
Detection Heads				
Head-P3	Detect cls/box/obj	80×80×256	80×80×(C+5)	SiLU
Head-P4	Detect cls/box/obj	40×40×512	40×40×(C+5)	SiLU
Head-P5	Detect cls/box/obj	20×20×1024	20×20×(C+5)	SiLU

3 Proposed System

The proposed system is thoroughly described within Fig. 1, which presents a comprehensive overview of its necessary components and design. Additionally, further insights into its structural architecture can be found detailed in Table 2. It integrated two modules involving *RGCA* and a *DCA* to effectively capture local and global contextual representation within strawberry leaf images. This integrated strategy addresses the challenges of accurate and real-time disease identification.

3.1 Overview

The proposed system combines channel guided attention and contextual aggregator based feature extraction to get optimal features. Later these mature descriptors is forward to YOLO-NAS to perform the training and test in real-time agricultural farming using robotic arm. Our features extracted modules are deviates from the standard channel attention and transformer network due to the inherent positional awareness embedded within the deep features through convolution and pooling operations. The

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process begins with normalizing the position-aware embedded patches generated from the channel output. Then, inside a high-dimensional space (α), multi-head attention sub-layers project queries and keys into various sub-spaces ($\alpha_1, \alpha_2, ...\alpha_h$). It allows the attention function to be executed in parallel over many sub-spaces, collecting a variety of feature correlations while avoiding over-fitting. The resulting attention features from each subspace are then merged to provide a richer representation. Subsequently, non-linear transformations are introduced by a Multi-Layer Perceptron (MLP) sub-layer. Two linear connections with ReLU activation functions make up this sub-layer, which efficiently processes the attention's subspace properties and improves the expressiveness of the model. Layer normalization and residual connections are features of both the MLP and the multi-head attention sub-layers. By stabilizing intermediate activations, layer normalization helps save over-fitting and training time. Residual connections, on the other hand, facilitate gradient flow in deeper networks, alleviating the vanishing gradient problem. This refined feature representation, obtained through the Transformer encoder, forms a strong foundation for subsequent tasks like feature fusion and detection within the overall architecture. The standard transformer encoder is shown in Fig. 2.

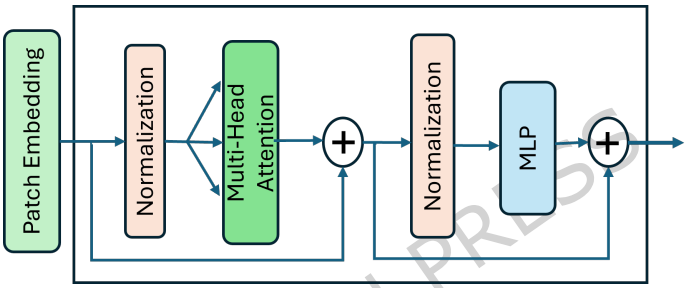


Fig. 2 A standard transformer encoder structure. Diagram illustrating the components of a transformer encoder block, including normalization, multi-head self-attention, and a multilayer perceptron.

The SLDI system has four key components, enable to accurately and efficiently identify diseases in real-time.

- Dual encoder, *RGCA* and *DCA* modules are used for the optimal feature extraction of the input image.
- *MSFF* and *PAN* modules are employed for feature fusion network.
- YOLO-NAS head network processes the fused features for precise disease region detection.
- Real-time implementation is accomplished by utilizing a robotic arm.

To increase the number of training samples, we used different data augmentation methods. Incorporating an efficient *RGCA* improves ResNet-101's capacity to extract unique features during the feature extraction phase. To do this, the *RGCA* concentrates on the object of interest rather than the background of the image. CNN's capacity to handle such input is enhanced by aggregating global feature information

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using a transformer encoder and a global self-attention mechanism. *MSFF* enhances the model's performance to recognize strawberry leaf diseases by capturing complex associated characteristics. The detection layers then analyze multi-scale feature maps to anticipate disease categories and create bounding boxes. These crucial details help farmers in identifying and finding these diseases. The upcoming subsections describe the technical details of individual components involve in our proposed system.

3.2 RGCA Encoded Module (M_1)

The proposed SLDI framework adopts ResNet-101 as its primary feature extractor to capture rich and discriminative representations from input images. Compared to the original ResNet down-sampling block, this network has the same number of parameters but improves overall network performance by minimizing feature information decrease on the feature map.

We incorporate the *RGCA* into the ResNet-101 architecture for enhanced feature extraction and expand the network's ability to capture essential features. The Global Average Pooling (GAP) and the Squeeze-and-Excite (SE) method are used for condensing Global spatial information into channel-specific descriptors. The SE approach uses two completely Fully Connected (FC) layers to emphasize inter-channel interactions and uses feature vectors to assess each channel's relevance. Mathematically, the SE process can be expressed as:

$$S = \sigma(f_{w1,w2}(y)) = W_2 ReLU(W_y), \tag{1}$$

where the channel information descriptor acquired by GAP is represented by y, the sigmoid activation σ, and the fully linked layers are represented by $f_{w1,w2}(y)$.

$$y = g(X) = \frac{1}{HxW} \sum_{i=1}^{H} \sum_{j=1}^{W} \tag{2}$$

The two-dimensional feature map, denoted by y, has dimensions of W and H. Additionally, the parameters of the two fully connected layers are $2C^2$.

SE's decay rate decreases the number of parameters to $2C^2/_r$. The *RGCA* block compresses the input feature tensor $X \in \mathbb{R}^{C \times H \times W}$ via a channel-wise GAP operation. Here, *(C- the number of channels, H -height, and W -width)* of a feature map. With this method, every spatial feature map of size $H - height \times W - width$ in the cth channel is reduced to a single weight coefficient, as seen below:

$$y_c = F_{sq}(X_c) = g(X_c), \tag{3}$$

Regarding channel dependencies, the Squeeze-and-Excitation (SE) model uses fully connected layers, whereas the *RGCA* block takes a different approach centered on capturing local cross-channel interactions inside a feature map. It is accomplished by applying a channel-wise GAP operation represented by $F_.$.

$$w_i = \sigma(\sum_{j=1}^{k} w^i y^j), y^j{}_i \epsilon \Omega^k{}_i, \tag{4}$$

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where the value of the k channel close to y_i is indicated by $\Omega^k{}_i$ and y_i is the descriptor of the ith channel.

These cross-channel interactions are efficiently performed by a rapid 1D convolution with a kernel size of k, which establishes the range of local cross-channel connections. The learned channel descriptors $S1$ of the *RGCA* are then multiplied by the input feature map to create a scaled feature map that shows the relative relevance of each channel. The following equation expresses C^{th} channel feature map:

$$\tilde{X}_c = F_{scale}(X_c, S_c) = X_c S_c, \tag{5}$$

where the weighted feature maps $\tilde{X}_1$,...., $\tilde{X}_c$,.... $\tilde{X}_C$ are expressed as $\tilde{X}_c$ and the channel's c^{th} weight coefficient is represented by S_c.

3.3 DCA Encoded Module (M_2)

The *DCA* is employed after feature extraction inspired by ResNet-101 backbone to improve the semantic understanding of the extracted features. The channel-wise attention are powerful at capturing local patterns, they often practice with modeling long-range dependencies due to their limited feature maps. To address this limitation, the *DCA* module leverages a *DCA* structure, that inherently captures global context and multi-scale features through self-attention mechanisms. By integrating this transformer-based module, the *DCA* refines the feature descriptors achieved from the CNN, allowing the network to understand the relationships between distant areas in the image. Furthermore, it improve context aggregation ultimately boosts the network's performance on tasks that require a deeper understanding of the entire scene or complex patterns that span across large spatial regions.

Once the feature map $F \in \mathbb{R}^{C\times H\times W}$ is extracted from the backbone model, it undergoes a flattening operation. Specifically, the spatial feature dimensions H and W are collapsed into a single vector, resulting in $N = H \times W$ patches. Each patch (originally a C-dimensional vector) is projected via a learnable linear layer to a new embedding space of dimension d, producing:

$$X = \text{Flatten}(F) \in \mathbb{R}^{N\times d} \tag{6}$$

where $N = H \times W$ is the number of image patches, and d is the embedding dimension.

The sequence X is forwarded through a Multi-Head Self-Attention (MHSA) module. First, the input tensor X is projected to Query (Q), Key (K), and Value (V) matrices:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \tag{7}$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d\times d_k}$ are learnable weight matrices, and d_k is the dimension per head. The attention output is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{8}$$

For multi-head attention, resultant from all heads are fused and projected:

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$$\text{MHSA}(X) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)W_O \tag{9}$$

where $W_O \in \mathbb{R}^{hd_k \times d}$ is a learnable projection matrix.

After the attention block, a position-wise Feed-Forward Network (FFN) processes the features:

$$\text{FFN}(x) = W_2\,\sigma(W_1 x + b_1) + b_2 \tag{10}$$

where $W_1 \in \mathbb{R}^{d \times d_{\text{ff}}}$, $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$, b_1 and b_2 are bias terms, and σ is an activation function, typically GELU or ReLU.

Residual connections are employed around both the MHSA and FFN blocks followed by normalization layer to stabilize training:

$$X' = \text{LayerNorm}(X + \text{MHSA}(X)) \tag{11}$$

$$X'' = \text{LayerNorm}(X' + \text{FFN}(X')) \tag{12}$$

Finally, the output $X'' \in \mathbb{R}^{N \times d}$ is reshaped back to spatial dimensions (H, W) with d channels and fused with the original CNN feature maps for further processing.

3.4 Multi-Scale Feature fusion (M_3)

The feature extraction network produces the disease characteristics, which then extracts them from processed pictures and displays them as various feature maps (abbreviated as C_2, C_3, C_4, C_5). Generally, the shallow feature map produced during the first stage of the feature extraction network has less impact on target categorization and provides little semantic information about the target. However, the broader receptive area represented by the feature values of individual pixels in the output feature map makes it more difficult to discern small objects in the deep feature map as the network layers become deeper.

To address this issue, the *MSFF* approach adds cross-stage branches to the MSFF and PAN propagation channels. High-level semantic feature maps are built by FPN in different sizes using a top-down architecture and lateral connections. Due to the long travel from low-level feature maps to high-level feature maps in FPN caused by the CNN's numerous layers, target contour edges and other low-level characteristics are largely lost in the high-level feature map. PAN raises FPN by using a bottom-up strategy to mitigate this.

Our approach effectively integrates feature maps from different stages using the routes generated by FPN and PAN, improving feature expressiveness for tiny objects such as strawberry leaf diseases. PAN uses a hierarchical fusion mechanism in its bottom-up approach. N3 is created by downsampling the low-level feature map P2, combining it with P3, and then starting again.

This process is repeated by down-sampling N3 and fusing it with P4 to generate N4 and incorporating P2 into the creation of N4, the resulting feature map benefits from enhanced low-level information. Similarly, P3 is directly involved in the generation of N5, ensuring that low-level features are preserved and strengthened throughout the fusion process, as mentioned in Fig. 3.

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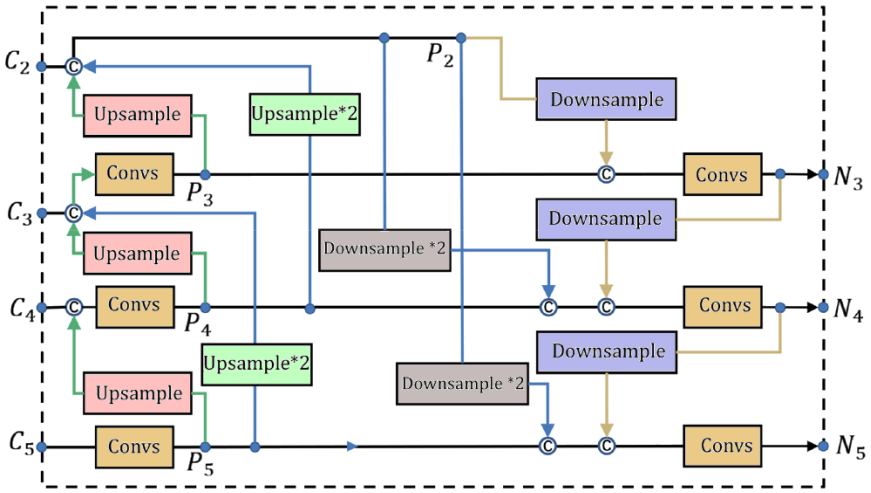


Fig. 3 Multi-scale features are fused using *MSFF*. FPN and PAN create two feature enhancement paths, forming a multi-branch structure to improve feature representation further.

First, in the *MSFF*, the feature map from the higher-level route is scaled using bi-linear interpolation to fit into the lower-level path dimensions of each cross-stage branch. Then, we repeatedly merge components of the scaled feature map with those of the lower-level route feature map. A single feature map with high-level and fine-grained features is produced because the information from several phases is integrated. A convolutional layer is appended at the end further to enhance the representational capacity of the fused feature map.

3.5 Strawberry Leaf Disease Detection (M_4)

The SLDI algorithm uses YOLO-NAS detection layers [1] for bounding box regression and category prediction. To predict the disease categories within each likely bounding box, it combines feature information and anchor settings with multi-scale feature maps from the *MSFF* stage. A multi-label classification procedure is then used, with the sigmoid function producing confidence scores for each prediction. Boxes with confidence ratings that surpass a predetermined threshold are further refined using non-maximum suppression. Lastly, the identified object boxes and disease classes are shown.

The incorporation of SLDI into agricultural production systems enables the quick diagnosis and monitoring of diseases, offering critical assistance to farmers. Strawberry leaf disease classes include powdery mildew leaf and fruit, leaf spot, gray mold, blossom blight, anthracnose fruit rot, angular leaf spot, and healthy leaf. These classes are visually represented in Fig. 4.

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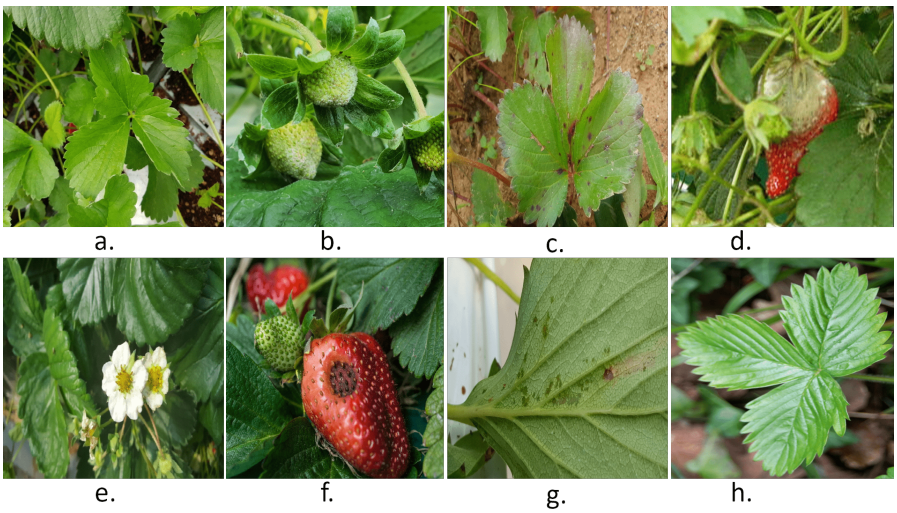


Fig. 4 Sample images of various strawberry diseases with class names. a.) powdery mildew leaf, b.) powdery mildew fruit, c.) leaf spot d.) gray mold e.) blossom blight f.) anthracnose fruit rot g.) angular leafspot and h.) healthy leaf

3.6 Robotic Inspection System Overview (M_5)

Although this study primarily focused on developing the SLDI algorithm for robust detection of strawberry leaf disease, our main research objective is to integrate this algorithm into a robotic platform to be used for robotic greenhouse inspection [38]. Three main components of this robotic system include a sensing system, a specialized end effector, and a robotic manipulator.

Fig. 5 illustrates the robotic strawberry plant disease detection system. A six degree of freedom (6-DoF) industrial manipulator (Universal Robot, UR5) is connected to the end-effector, which has an RGB-D camera (Intel Realsense 435i) installed on it. The sensor is positioned parallel to the end-effector. To simulate greenhouse conditions, the robotic manipulator is currently placed on a mobile platform near strawberry plants on a table.

The strawberry leaf disease inspection SLDI detection algorithm is optimized for deployment on the UR5 robot using the Tensor-RT format and UR-Cap module. Fig. 6 presents the real time robot based plant scanning system. To comprehensively scan the strawberry plant, three predefined way-points (1, 2, and 3) were established to cover a scanning zone. The RGB camera is mounted at the tip of the robot rend effector and moves in linear path for scanning the strawberry plants.

Fig 7 illustrates the robot joint position and velocity during the scanning process. The graph shows the results for 20 seconds. Fig 7 (a) illustrates the joint's position during motion from way-point 1 to 3. As shown in Fig 7 (b), the end effector's velocity is quite low, which contributes to effective scanning and accurate detection results. Likewise, the acceleration graph of the end effector remains within an acceptable range. The robot was equipped with a Jetson AGX Orin microcontroller to enable smooth

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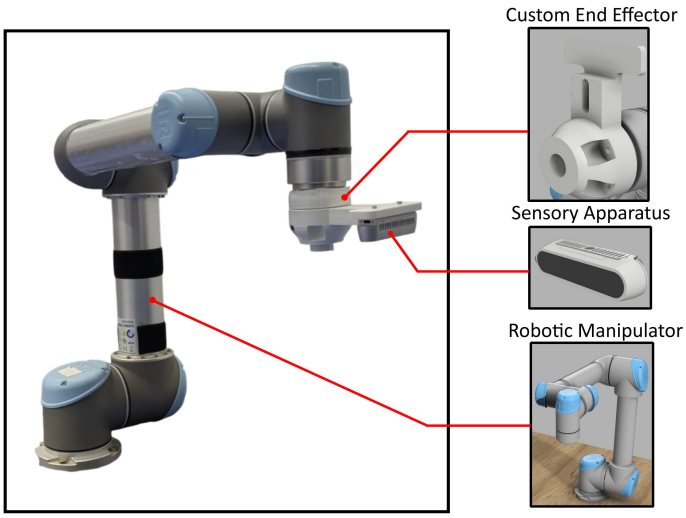


Fig. 5 The robotic disease detection system. Illustration showing the key elements of a robotic system: the robotic manipulator, a custom end effector, and a sensory apparatus.

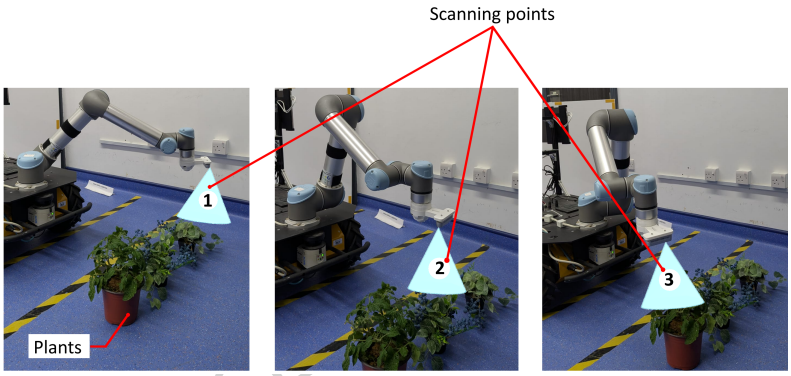


Fig. 6 The process of automated plant health monitoring using a robot to scan plants from various perspectives.

image capture and efficient leaf detection. The experiments were carried out in the Agri-Robotics Lab at the Khalifa University.

4 Experiments

This section provides a complete review of the proposed SLDI object identification method. We begin by outlining the experimental setup and datasets employed in this research. Next, we describe the SLDI implementation on the robotic arm UR5 [41]. The assessment process is then described, along with metrics (such as F1-score, precision, and recall), confusion matrix analysis, and loss function.

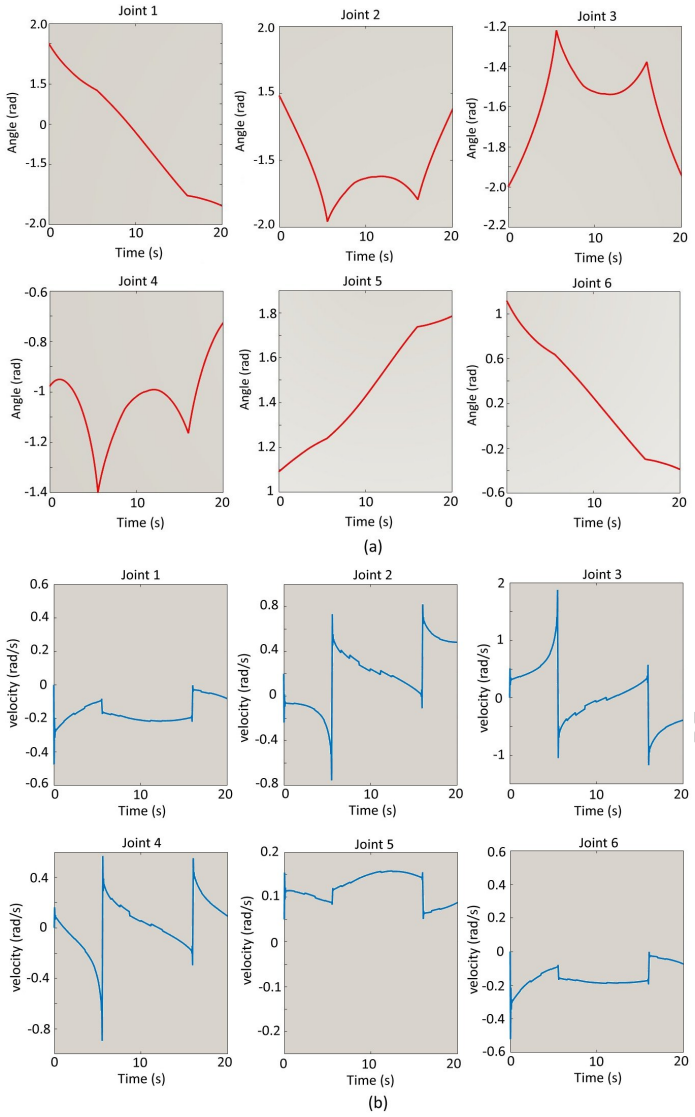


Fig. 7 Simulation results (a) Joints position, and (b) Joints velocity

A comprehensive evaluation of SLDI performance is conducted through a comparative analysis with seven prominent object detection models. This in-depth study provides substantial evidence of SLDI's superior performance compared to contemporary detection methods.

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4.1 Implementation Setup

The experimental setup utilized a high-performance computing system equipped with an NVIDIA RTX A6000 GPU, 64 GB of RAM, and an Intel Core i9 13th Generation CPU. This system was employed exclusively for the training phase of the proposed Strawberry Leaf Disease Inspection (SLDI) model. The model was implemented using the PyTorch deep learning framework, and training was conducted over 100 epochs in each experiment to ensure model convergence and stability. To evaluate computational efficiency, we processed input images with a resolution of 640 $\times$ 640 pixels using a batch size of 32. Under these conditions, the system achieved an average processing time of approximately 12.7 milliseconds per batch on a single NVIDIA RTX A6000 GPU. These results demonstrate the suitability of the proposed model for real-time training workflows, with sufficient capacity to handle high-resolution agricultural image data. The experimental outcomes further validate the SLDI model's effectiveness in accurately detecting and classifying various strawberry leaf diseases. The trained model was later implemented on UR5 Robotic Arm. The details are in the sub sequent sections.

4.2 Datasets

Datasets are fundamental to training and validating models for specific tasks. This subsection outlines the characteristics and purposes of the two datasets employed in this research. The datasets are: PlantVillage [15] and PlantDoc [40]. The class imbalance present in the two datasets is reduced by applying, different data augmentation and label smoothing approaches. To name few are the color transformation, Mosaic [5], Mixup [57], and Flip etc. For model training, the aforementioned datasets were divided into training set, validation set, and test set, in a 6:3:1 ratio respectively.

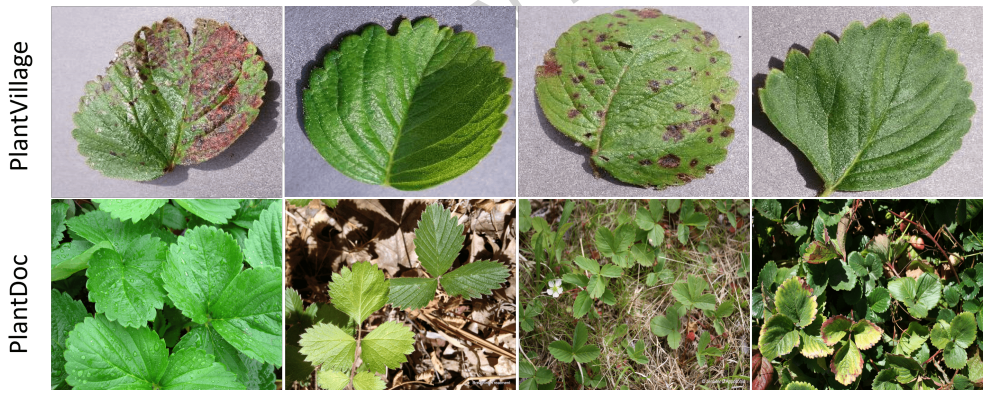


Fig. 8 Sample images for the study are obtained from two widely recognized datasets: the PlantVillage Dataset [15] and PlantDoc [40].

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4.2.1 Plant Village

We have used public PlantVillage dataset [15], a comprehensive collection of images capturing healthy and diseased strawberry leaves under controlled conditions. The PlantVillage dataset was primarily prepared by researchers at Penn State University (Pennsylvania State University) in the United States. From this dataset, some of the samples are presented in Fig. 8.

4.2.2 Plant Doc

The PlantDoc [40] contains a comprehensive collection of image data related to plant health and pathology. The PlantDoc dataset was prepared by researchers at the Indian Institute of Technology (IIT) Gandhinagar, India. The dataset includes a large variety of images capturing various aspects of plant foliage, ranging from healthy specimens to those infected with a variety of diseases. Fig. 8 presents some of the PlantDoc images. The comprehensive details regarding the classification of images used for training, validation, and testing purposes can be found expansively elaborated in Table 3.

Table 3 Statistics for strawberry subclasses in the PlantVillage and PlantDoc datasets.

Dataset	ID	Subclass / label	Train	Val	Test	Total
PlantVillage Dataset [15]						
	0	Leaf scorch spot	388	48	44	480
	1	Healthy leaf	392	96	95	583
PlantDoc Dataset [40]						
	0	Leaf scorch spot	288	90	70	548
	1	Healthy leaf	192	60	70	322
	0	Healthy leaf	200	50	50	300
	1	Angular leaf spot	235	70	60	365
	2	Anthracnose fruit rot	100	38	32	170

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4.3 Implementation on UR5 Robotic Arm

The proposed SLDI algorithm is implemented on a real UR5 robotic arm for real-time detection within a greenhouse environment. This implementation has been successfully tested in a laboratory environment, providing initial validation of its capabilities. Future work will focus on extensive field testing to assess its performance and reliability under real-world conditions. The model is converted into a suitable format (RT-Tensor) and programmed using URCap functionalities.

In this work, we focused on seven prevalent diseases: angular leaf spot, Anthracnose fruit rot, blossom blight, gray mold, leaf spot, powdery mildew fruit, and powdery mildew leaf. Fig. 9 illustrates the experimentation setup for detecting real-time strawberry diseases. The key components of the setup include:

- A 3D printed custom end effector, which is the tool or attachment mounted on the robotic arm.
- An Intel RealSense camera, which is mounted on the UR5 universal robot.
- The camera and end effector are integrated as part of the overall experimental setup to enable real-time monitoring and detection of strawberry diseases.
- The robot was equipped with a Jetson AGX Orin micro-controller to enable smooth image capture and efficient leaf detection.

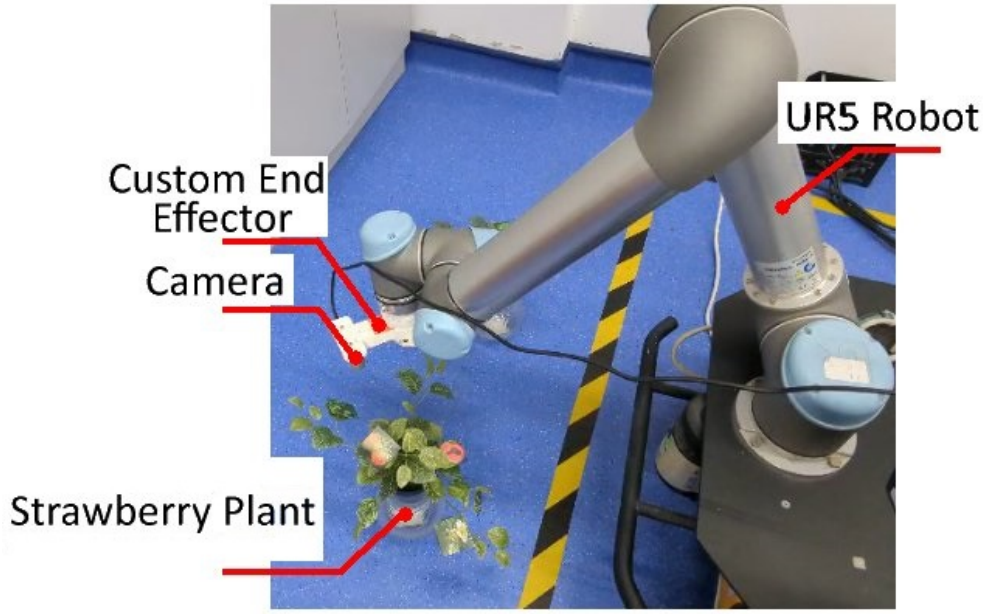


Fig. 9 Integrated robotic platform: A UR5 arm, custom end-of-arm tooling, and an RGB vision system for detailed observation.

4.4 Performance Metrics

The Mean Average Precision (mAP) statistic is often used to evaluate object-detecting algorithms. This measure is computed by comparing predicted bounding boxes to their corresponding ground truth bounding boxes. The Intersection Over Union (IoU) is an important component of this comparison since it evaluates the overlap between the anticipated bounding box and the ground truth.

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$$IoU = \frac{\text{area of overlap}}{\text{area of union}} \tag{13}$$

Precision is the proportion of true positives out of all the positive predictions made by a model. It measures how accurate the model's positive predictions are.

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{14}$$

Recall is the proportion of true positive samples that the model correctly identified as positive out of all the true positive samples. In other words, recall measures how well the model is able to find or detect the positive samples. A higher recall value indicates the model is able to correctly identify a larger proportion of the actual positive cases.

$$Recall = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{15}$$

AP^{IOU} is the average area under the precision-recall curve for a given IoU threshold. It provides a performance metric for a specific class or category. To get an overall measure of detection performance, mAP^{IOU} is calculated. mAP^{IOU} is the mean of the AP^{IOU} values across all classes in the dataset. This single metric represents the overall detecting performance of the object identification system

$$mAP^{IoU} = \frac{1}{N} \sum AP_n^{IoU} \tag{16}$$

where $n \; \epsilon \; class1, class2, ..., classN$

The F1 score combines these two factors into a single value that indicates how well the model can accurately predict outcomes. An improved ability to balance accurately detecting instances that are positive with reducing false positives is shown by a higher F1 score for the model. To summarise, the F1 score evaluates a classification model's overall efficacy by considering its precision (accuracy in producing positive predictions) and recall (ability to identify positive occurrences). It is a useful metric for evaluating how well a model can predict different classes.

$$F1 - score = 2 \times \frac{Recall \times Precision}{Recall + Precision} \tag{17}$$

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5 Ablation Studies

In this section, a comprehensive evaluation of the experimental results is presented. Both qualitative and quantitative assessments are conducted, showcasing the performance of each SLDI through a series of key metrics, including precision (P), recall (R), mAP, and Frames Per Second (FPS). Fig. 10 presents the real-time results of strawberry leaf disease detection. The proposed model effectively identifies diseases in strawberry plants under laboratory lighting conditions, which differ from those in the training dataset. The proposed SLD is evaluated over testing samples taken from the Plant Doc and PlantVillage dataset. The confusion matrix for both datasets are given in Figure 11.

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Fig. 10 Robotic system detects strawberry diseases in real-time with SLDI Integration.

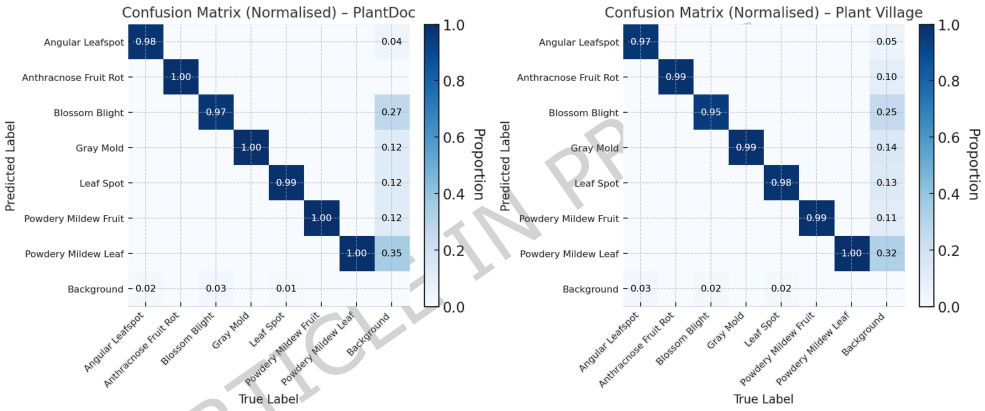


Fig. 11 Confusion matrix for both datasets over testing samples.

5.1 Quantitative Assessment of BB, RGCA, DCA, and MSFF

To verify the performance of the proposed system in terms of accuracy and speed of leaf disease detection, we conducted comprehensive experiments exploring the impact of three key mechanisms: the *RGCA, DCA, and MSFF* methods with baseline Backbone architecture. The experimental results, as summarized in Table 4, indicate that the SLDI, incorporating these sophisticated mechanisms, achieves substantial performance improvements over the baseline model. Specifically, while the unchanged Backbone yields an mAP@0.5 of 70.8%, the incorporation of the RGCA mechanism

Table 4 Performance analysis of various modules collection, where BB, ST, SE, represents ReseNet-50, SwinTransformer, and Squeeze-and-Excitation, while RGCA, DCA, and MSFF indicates our proposed system components. The best and runner up performance is denoted by Blue and Red.

Methods	FLOPs (G)	mAP@0.5 (%)	FPS	TP (M)
BB	109.0	70.8	110	44.8
ST	435.0	69.0	78	48.1
BB + SE	111.5	71.8	108	47.5
BB + RGCA	110.5	72.6	112	45.2
BB + MSFF	111.0	71.5	118	47.0
BB + DCA	129.0	71.3	98	82.0
BB + RGCA + MSFF	112.0	72.8	85	47.3
BB + MSFF + DCA	132.0	72.9	80	85.0
BB + RGCA + MSFF + DCA	132.5	73.6	81	85.1

results in a 1.8% improvement (increasing the mAP to 72.6%), the addition of the DCA component provides a 0.5% enhancement (reaching 71.3%), and the integration of the MSFF method improves the mAP by 0.7% (yielding 71.5%). Most notably, the optimal configuration of SLDI, which synergistically combines the RGCA, MSFF, and DCA mechanisms, attains the highest accuracy of 73.6% while maintaining an efficient inference speed of 81 FPS. The second-best performance, achieved by the Backbone + MSFF + DCA configuration, reaches an accuracy of 72.9% at 80 FPS. Although these architectural enhancements lead to a modest increase in computational complexity from 109.0G FLOPs to 132.5G FLOPs the significant gains in detection accuracy justify this trade-off, thereby demonstrating the practical utility and effectiveness of the proposed design in real-world leaf disease detection applications.

The Fig.12 illustrates the AP scores of seven evaluated methods on the Plant Village and Plant Doc datasets.

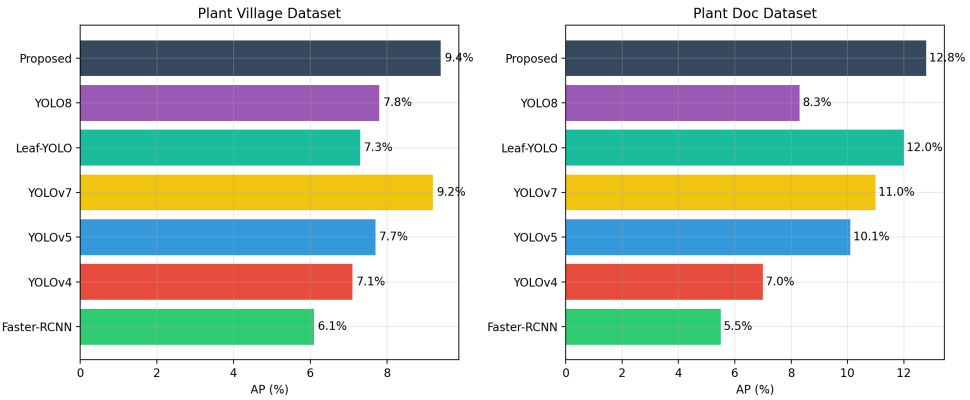


Fig. 12 Average precision performance of various detection models using Plant Village dataset.

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Notably, the results clearly demonstrate that the proposed model, incorporating the ECA mechanism, Transformer Encoder, and CSFF strategy into ResNet-50-D, outperforms existing baselines including Faster R-CNN, YOLOv4, YOLOv5, YOLOv7, LeafYOLO, and YOLOv8s, especially in detecting small and challenging plant disease instances. Specifically, on the Plant Village dataset, the proposed model achieves an AP of 9.4%, outperforming YOLOv8s (7.8%) and YOLOv7s (9.2%). Similarly, on the more complex Plant Doc dataset, the proposed model achieves an AP of 12.8%, surpassing YOLOv8s (8.3%) and LeafYOLO (12.0%). Compared to Faster R-CNN, the proposed method records significant improvements of 3.3% and 7.3% on the Plant Village and Plant Doc datasets, respectively. These results validate the effectiveness of the proposed enhancements in improving detection accuracy, particularly in scenarios involving limited data or smaller object sizes.

5.2 Comparison with popular Object Detection Methods

We evaluated the performance of the proposed Strawberry Leaf Disease Inspection (SLDI) model against several cutting-edge object detection techniques, including Cascade R-CNN [8], SSD [24], YOLOv3 [35], YOLOv4 [5], YOLOv5 [17], YOLOv7 [44], YOLOv8s [16], and YOLOv9 [49]. As illustrated in Fig.13, the comparison is based on widely accepted object detection metrics, including mAP@0.5, mAP@0.5:0.95, precision, recall, and inference speed (frames per second, (FPS)). Fig. 13 and Table 5 present the comparative performance of these models on the PlantVillage and PlantDoc datasets. SLDI achieved the highest mAP@0.5 of **89.40%**, outperforming all other

Table 5 Comparison of recent strawberry-leaf disease detection models. Best and runner-up scores are highlighted in and red, respectively.

Model	mAP@ 0.5	mAP@ 0.5:0.95	Precision	Recall	FPS	Params (M)	FLOPs (G)
Cascade R-CNN	66.30	41.50	75.50	74.00	26.50	84.0	708
SSD	52.80	32.10	70.40	71.20	62.80	34.3	45
YOLOv3	65.40	38.90	74.20	72.20	76.00	61.5	65
YOLOv4	71.30	44.80	83.00	81.10	71.80	64.4	96
YOLOv5	72.40	45.90	72.10	71.10	74.60	12.6	16
YOLOv7	73.20	46.70	73.00	72.00	70.50	37.2	105
YOLOv8-s	71.80	45.10	71.60	71.00	76.20	11.2	28
YOLOv9	88.10	60.40	87.20	85.00	74.30	10.0	25
SLDI	**89.40**	**62.80**	**91.10**	**88.50**	**76.50**	**9.8**	**22**

models. It also attained the top precision of **91.10%** and a recall of **88.50%**, demonstrating its ability to identify strawberry leaf diseases with high accuracy and low false-positive rates. While YOLOv8s achieved the best FPS (**76.20**), SLDI delivered a comparable speed of **76.50 FPS**, making it suitable for real-time deployment.

Moreover, the SLDI model reached an impressive apex in performance, attaining the maximum mean Average Precision (mAP) within the IoU threshold range of 0.5 to 0.95 with a notable score of **62.80%**. This outcome underscores its robust

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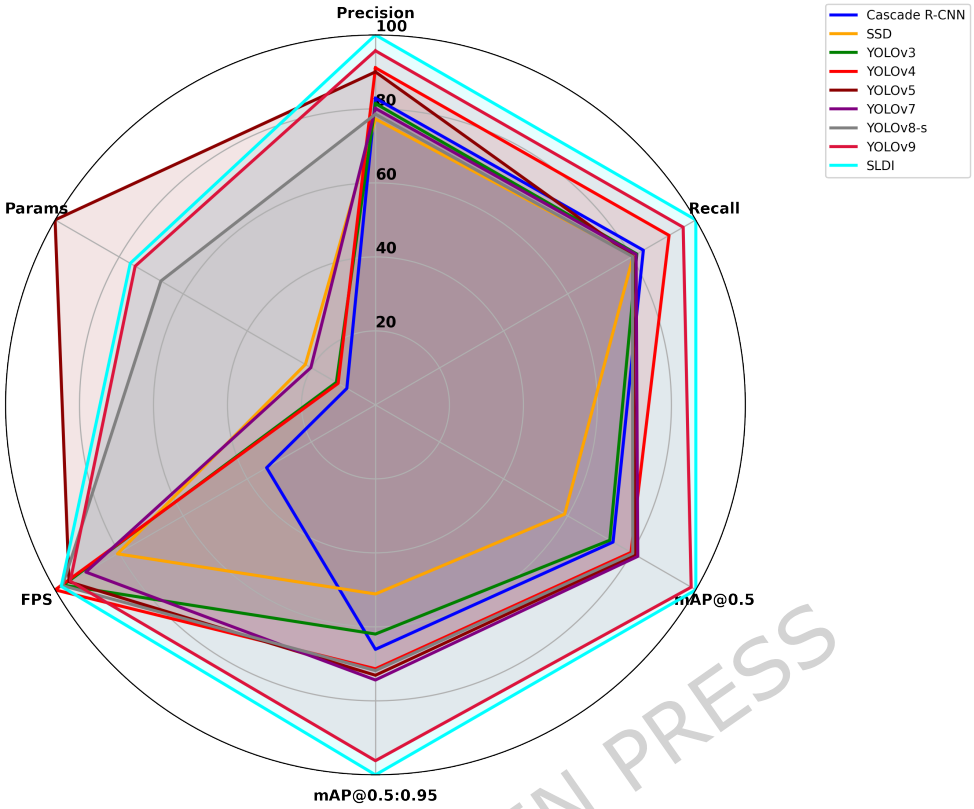


Fig. 13 We compared several state-of-the-art (SOTA) models on various datasets using performance measures such as speed (FPS), accuracy, recall, and mAP@0.5.

generalizability across a range of Intersection over Union (IoU) thresholds, a critical factor in evaluating detection accuracy. Furthermore, the model is fundamentally optimized through its compact architecture, which consists of merely **9.8 million parameters**. Additionally, it operates with a computational requirement of **22 Giga Floating Point Operations per second (GFLOPs)**, emphasizing its exceptional performance. Such characteristics make it ideally suited for a wide range of practical implementations, where resource constraints are a substantial consideration.

Fig. 14 shows the inference results on unseen test images, confirming SLDI's robustness in detecting and localizing strawberry leaf diseases under diverse visual conditions.

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Fig. 14 Real-time evaluation on unseen images: comparison of the proposed detector with leading object-detection baselines.

Overall, the proposed method consistently outperforms existing models in both detection accuracy and efficiency, making it a promising solution for real-time plant disease monitoring.

6 Discussion

Crop diseases are serious risks to agriculture due to their detrimental impacts and rapid spread, resulting in large losses in productivity and quality. Modern technologies are needed to address these issues and provide quick and reliable identification, which is a critical responsibility in agriculture. Prompt detection and identification enable farmers to take appropriate action, reducing losses and maximizing resource utilization.

Our proposed SLDI method utilized in this study significantly increases real-time disease diagnosis. Our suggested model surpasses current cutting-edge algorithms by establishing a remarkable balance between detection accuracy and speed. However, given the diversity of disease species and the apparent resemblance in colour and texture of certain leaf diseases, there is room to increase overall detection accuracy.

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The proposed strawberry leaf disease inspection model is integrated into a robotic system for real-time disease inspection. The experiments are conducted in a controlled lab environment with infected leaves of strawberry plants. The robotic system, equipped with the detection model, demonstrates high efficiency in detecting strawberry diseases in real-time. However, the study has some limitations. The performance of the model has only been evaluated under controlled laboratory conditions, which may not fully represent real-world field environments. Variations in lighting, background noise, leaf orientation, and natural disease progression in outdoor settings could affect detection accuracy. Additionally, the dataset used may not cover all possible disease types or stages, potentially limiting the model's generalizability to unseen cases.

Further more, Our long-term research objective is to implement the strawberry leaf disease inspection (SLDI) on IoT devices as well to provide ongoing disease monitoring in greenhouses. For example, SLDI can work with an integrated IoT based system to let farmers know when the disease detection reaches a certain level so they can take preventive measures to reduce crop losses. Notably, the number of model parameters is strictly limited when the model is integrated into hardware devices. To preserve the model's detection performance, we concentrate on parameter optimization in our upcoming research phase. In agricultural contexts, striking this balance is essential for practical implementation, especially regarding enhancing monitoring accuracy.

7 Conclusion

This study presents SLDI, a novel disease identification method that consists of two parallel encoders: Receptive Guided Channel Attention (RGCA) and a Deep Context Aggregator (DCA). The *RGCA* enhances the representation of spatial features by focusing on channel-wise attention, while the *DCA* captures long-range dependencies and contextual information within the image. These complementary feature sets are then fused using a dedicated Feature Fusion Network. The fused features are subsequently processed by a prediction head to perform the final output. The evaluations carried out on the PlantVillage and PlantDoc datasets confirm the ability of SLDI to outperform existing methods, achieving a precision of 91.10%, a recall of 88.50%, and a real-time frame rate of 76.50 fps.

To facilitate the transition from research to practical application, SLDI has been implemented on a robot-based system for real-time disease detection. Preliminary tests within a lab controlled environment have demonstrated promising outcomes. Future research endeavors will concentrate on optimizing the system for deployment in greenhouse environments, incorporating additional disease datasets of other crops, and investigating advanced robotic manipulation techniques to further enhance disease management practices. The successful implementation of SLDI has the potential to significantly increase disease control strategies and contribute to the advancement of sustainable agriculture.

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Declarations

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

- **Ethics approval:** This study did not involve human participants or animals, thus no ethics approval or consent to participate was required.
- **Consent to participate Consent for publication:** All authors have provided their consent for publication.
- **Availability of data and materials:** This study analyzed a combination of publicly available datasets and data collected by the authors. The publicly available datasets can be accessed at [www.plantvillage.org] and [https://github.com/pratikkayal/PlantDoc-Dataset]. If you want to request the complete dataset, please email the corresponding author.
- **Competing Interests:** The authors declare no competing interests.
- **Funding:** Not applicable
- **Authors' contributions:** A.K: As the first author, responsible for the main aspects of the research, including experimental design, data collection, analysis and interpretation, as well as drafting the initial manuscript. Conceptualization, methodology, software, writing-original draft, formal analysis, writing-reviews and editing. R.S: Robotic setup, implementation and result analysis, Reviewing and Editing the draft. S.U.K: Reviewing and Editing, Visualization, and Validation. I.H: As the principal corresponding author, responsible for the overall guidance of the research, team coordination, resources, final review, and correspondence with the journal.
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- **Abbreviations:**
  - SLDI: Strawberry Leaf Disease Inspection
  - IoT: Internet of Things
  - RGCA: Receptive Guided Channel Attention
  - DCA: Deep Context Aggregator
  - MSFF: Multi-Scale Feature Fusion
  - RGB:Red Green Blue
  - ML: Machine Learning
  - GNN: Graph Neural Networks
  - CNN: Convolutional Neural Networks
  - ViT: Vision Transformers
  - BR: Binary Relevance
  - AFormer: Agricultural Aerial Transformer
  - FFLA: Feature-Fuse Local Attention
  - PMVT: plant-based MobileViT
  - YOLO-NAS: You Only Look Once-Neural Architecture Search
  - MLP: Multi-Layer Perceptron
  - PAN: Pyramid Attention Network
  - GAP: Global Average Pooling

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– SE: Squeeze-and-Excite
– FC: Fully Connected
– Q, K and V:Query, Key and Value
– FFN: Feed Forward Network
– FPN: Feature Pyramid Networks
– RGB-D: Red Green Blue-Depth
– DoF: Degree of Freedom
– UR5: Universal Robot 5
– mAP: Mean Average Precision
– IoU: Intersection Over Union
– FPS: Frame Per Seconds

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